

Covariance-Method Claims Are Ensemble-Relative: Named Measures and an Audit Program

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Abstract

A claim about the behavior of a covariance or correlation method is a claim about a distribution over matrices, and the distribution is rarely stated. Papers validate estimators, factorizations, and portfolio constructions on whatever generator is at hand: a Wishart draw, a uniform LKJ sample, one empirical panel. We measure how much the choice matters. Across fifteen named, seeded ensembles at identical dimension and sample size, the Ledoit–Wolf shrinkage intensity ranges from 0.07 to 0.86, and eleven audited claims from eleven fields, from hierarchical risk parity to the Mantel test, each flip verdict as the ensemble changes, for stated structural reasons. The results assemble into a matrix of claims against measures whose outlined cells mark the structure each claim was demonstrated on or intended for. The `randomcov` library names the measures and seeds the draws, so a published claim can be re-run under all fifteen and its verdict reported per measure. A joint leaderboard of the scikit-learn and `precise` covariance estimators has five different winners across the fifteen ensembles. Eleven audits are completed here; a docket of candidates from astronomy, genomics, neuroscience, and chemometrics, including provably ensemble-invariant claims that serve as null controls, is assembled for the rest.

1 The unstated measure

Empirical claims about covariance methods have the form “estimator A beats estimator B ,” “the shrinkage intensity is typically moderate,” “the factorization recovers structure,” with performance averaged over random test matrices. The average is taken with respect to some generating law, and the law is usually inherited from convenience, and from the field. Finance validates on equity panels; econophysics on the same panels with random-matrix nulls; geo-statistics on smooth kernels; psychometrics on low-rank factor models; graphical modeling on sparse precisions; cosmology on mock catalogs of ordered correlation-function bins.

Different laws produce matrices with different spectra, different locality, and different factor content. A verdict certified under one law carries no warrant under another. The measurements below show that the omission is not a technicality.

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2 Precedents

That method comparisons inherit their verdicts from the simulation design is established in several literatures. Morris et al. [2019] require simulation studies to declare the data-generating mechanism as a design choice of equal rank with the estimand. Boulesteix et al. [2013] trace the over-optimism of self-reported comparisons to test beds chosen by the method’s own authors. Nießl et al. [2022] re-analyze one benchmark under hundreds of design and analysis options and find that almost any method can reach almost any rank. Hooker [1995] made the complaint for heuristics; Wolpert and Macready [1997] give the limit statement, that averaged over all objectives all optimizers tie.

The correlation-matrix version also has precedent. Joe [2006] and Lewandowski et al. [2009] supply the standard samplers. Hardin et al. [2013] argue that the habitual generators produce unrealistic matrices and bias the evaluation of statistical techniques, and propose a better one. Marti [2020] learns a generator from financial correlation matrices and reports, in accompanying experiments, that the HRP advantage fades on matrices without hierarchical structure.¹ Papenbrock et al. [2021] evolve synthetic correlations to ask which market regimes favor which allocator. Alvarez et al. [2014] document the Bayesian mirror image: posterior conclusions about a covariance track the prior family placed on it.

The present paper adds the instrument and the audits: a fixed battery of fifteen named, seeded, public measures; the re-execution of eleven published claims from eleven fields under all of them; the claims-by-measures matrix as the reporting format; and provably invariant theorems run as null controls.

3 Fifteen named ensembles

`randomcov` generates correlation matrices under named, seeded measures: `lkj`, `onion`, `vine` (elliptope samplers; Joe, 2006, Lewandowski et al., 2009); `wishart`, `spectrum` (spectral laws, including Marchenko–Pastur and spiked); `factor`, `hierarchical`, `block.equicorr` (low-rank and clustered structure); `ar1`, `kernel` (banded and locality-driven fields); `sparse.precision` (graphical-model structure); `archakov.hansen` (log-matrix parameterization; Archakov and Hansen, 2021); `residuals`, `walk`, `animals` (empirical-flavored and trajectory ensembles).

Variance vectors are drawn separately, so correlation structure and scale can be crossed. Every generator accepts an explicit seed, and every number in this paper descends from one. The scripts sit beside the paper.

4 Demonstration: one estimator, twelve-fold spread

The Ledoit–Wolf shrinkage intensity [Ledoit and Wolf, 2004] is a workhorse diagnostic: how far the sample covariance is pulled toward its target. Fix $n = 30$ variables, $T = 60$ observations, twenty replications, and vary only the generating ensemble.

¹gmarti.gitlab.io/qfin/2019/12/04/hierarchical-risk-parity-part-3.html

ensemble	intensity	sd
animals	0.072	0.013
residuals	0.096	0.012
factor	0.101	0.038
archakov_hansen	0.120	0.022
kernel	0.132	0.082
walk	0.199	0.038
block_equicorr	0.247	0.096
lkj	0.346	0.031
vine	0.355	0.022
onion	0.363	0.030
hierarchical	0.369	0.070
ar1	0.469	0.226
wishart	0.506	0.032
spectrum	0.532	0.058
sparse_precision	0.862	0.066

The intensity spans 0.072 to 0.862 at identical (n, T) . A paper reporting “typical” shrinkage from one generator is reporting a property of its generator. The ordering is interpretable: strong low-rank structure (**animals**, **residuals**, **factor**) wants little shrinkage, and near-unstructured draws (**wishart**, **sparse_precision**) want a great deal. The within-ensemble standard deviation is itself informative: **ar1** spans 0.23 across its own parameter draws, so even a named ensemble can hide a range.

5 Demonstration: one algorithm, one failing ensemble

A deterministic algorithm for multinomial probit choice probabilities (the **winning** package; paper in that repository) reports median absolute error by ensemble after fitting a structured covariance grammar. Under twelve of the fifteen measures the error is $1\text{--}4 \times 10^{-5}$; under dense-strong ensembles it is $1\text{--}4 \times 10^{-4}$; under **kernel** fields (Matérn and RBF) it is 4×10^{-3} . The failure has a mechanism: kernel covariance is locality, approximately Markov, and offers a low-rank factorization no good purchase. A validation run on any single ensemble would have reported either “essentially exact” or “inadequate,” both correctly, for the same method.

6 Eleven audits, eleven fields

The audit protocol is short: take a published claim, re-run its demonstration seeded under every named ensemble, report the verdict per ensemble, and flag the flips. We audit one claim from each of eleven fields, seven about estimators and constructions (Table 1) and four about inference and dimensionality (Table 2). Figure 1 assembles the results into a single matrix and outlines each claim’s *home turf*, the correlation structure it was demonstrated on or intended for. All runs use $n = 30$ with twenty replications unless stated; sampled panels use $T = 60$ except where the claim’s own regime dictates otherwise.

Finance. López de Prado [2016] reports that hierarchical risk parity delivers lower out-of-sample variance than quadratic minimum variance, and names estimation error in the inverse as the culprit. We build both portfolios from the same sample covariance and score the true

variance $w^\top \Sigma w$ under the generating covariance. Column 1 of Table 1 gives the median ratio of HRP to long-only minimum variance. HRP wins on every draw under `sparse_precision` and splits draws roughly evenly under `ar1` and `hierarchical`, the ensembles whose dependency structure the bisection tree can match; it loses by an order of magnitude or more under `animals`, `factor`, and `archakov_hansen`, where the true matrix rewards positions a long-only tree cannot take. Against unconstrained minimum variance the losses grow by many more orders of magnitude: under `walk` the median ratio is $\sim 10^{8.7}$, because near-singular ensembles admit near-perfect hedges that only signed weights reach.

Econophysics. Laloux et al. [1999] flatten the eigenvalues of the sample correlation that fall below the Marchenko–Pastur edge $(1 + \sqrt{n/T})^2$ and report a cleaner matrix. Column 2 gives the Frobenius-loss ratio of clipped to raw sample correlation. Cleaning is excellent under `sparse_precision` (0.30) and `hierarchical` (0.76), where the true spectrum really is spikes over a bulk. It damages the estimate under eleven of fifteen ensembles, worst, at 1.43–1.47, under `animals`, `residuals`, `kernel`, and `archakov_hansen`, where genuine eigenvalue mass lives below the edge and is erased as noise. The recipe presumes signal means spikes; locality puts signal in the bulk.

Geostatistics. The screening effect of Stein [2002] says that a few nearest neighbors, here the most correlated, nearly attain the full kriging variance. This is a population property, so no sampling is involved. Column 3 reports the median ratio of conditional variance given the best five predictors to conditional variance given all $n - 1$. Screening holds essentially exactly under `ar1` (Markov, ratio 1.00), `sparse_precision`, `hierarchical`, and `block_equicorr`. It fails several-fold under `onion` and `vine`, and under `kernel` itself (5.7): the ensemble mixes Matérn with squared-exponential fields, and for the smooth draws interpolation is nearly deterministic from all sites but not from five. Stein’s own caveat, that screening depends on smoothness, reappears inside the one ensemble geostatistics would call home.

Psychometrics. Horn [1965] retains the components whose sample eigenvalues exceed the corresponding 95th-percentile eigenvalue of independent data of the same shape; parallel analysis is the standard answer to “how many factors?” Column 4 gives the mean retained count: 1.8 under `sparse_precision` to 7.3 under `onion` at identical (n, T) . The same sweep scores the eigenvalue-greater-than-one rule of Kaiser [1960] against the population count of eigenvalues above one. Kaiser over-extracts under `factor` (5.7 retained against 3.0 true), under-extracts under `wishart` (11.2 against 13.1), and disagrees with parallel analysis by a factor that is itself ensemble-dependent: 1.2 under `residuals`, 7 under `sparse_precision`. The number of factors a study reports is, in part, a property of the generating measure its data resemble.

Data assimilation. Ensemble Kalman filtering runs member-starved, and the standard repair is covariance localization: taper the sample covariance entrywise with a Gaspari–Cohn function [Gaspari and Cohn, 1999] of inter-variable distance [Houtekamer and Mitchell, 2001, Hamill et al., 2001]. Paz and Sánchez [2015] arrived at the same recipe for correlation-function bins as covariance tapering, so this audit serves two literatures at once. With $N = 20$ members for $n = 30$ variables and a taper of index distance with half-radius 6, column 5 reports the Frobenius-loss ratio of tapered to raw. Localization helps wherever the estimate is noisy and the truth carries no long-range structure in taper coordinates, including the dense ellipsope

samplers, where variance reduction beats the bias; it hurts under **factor**, **residuals**, and **animals** (1.66–1.71), whose common factors are long-range by construction. It also hurts under **kernel** (1.46), the one ensemble that genuinely is local, because kernel locality lives in a latent two-dimensional point cloud rather than in index order. Localization requires locality in the coordinates the taper can see.

Signal processing. Diagonal loading is the standard repair for the sample-covariance MVDR beamformer in the snapshot-limited regime [Cox et al., 1987, Carlson, 1988]. With $T = 40$ snapshots, a random unit steering vector, the distortionless constraint, and loading $\delta = 0.1 \text{tr}(\hat{\Sigma})/n$, column 6 reports the median ratio of true output power, loaded over unloaded. Loading halves the output power under half the ensembles. It multiplies the power under **archakov_hansen** (21), **animals** (33), **kernel** (136), and **walk** ($10^{7.9}$). The mechanism is the hedge from the finance audit seen from the other side: under near-singular ensembles the beamformer’s null-steering lives in the smallest eigenvalues, which is exactly what loading erases.

Graphical modeling. Friedman et al. [2008] attach to the graphical lasso the promise that a sparsity prior on the precision matrix improves estimation. The same promise underwrites gene-network inference [Schäfer and Strimmer, 2005], partial-correlation connectomes [Smith et al., 2011], and sparse cosmological precision matrices [Padmanabhan et al., 2016]. Column 7 reports the Frobenius loss of the cross-validated glasso precision against the true precision, relative to inverting a Ledoit–Wolf shrinkage estimate. The sparsity prior wins clearly where precisions are genuinely sparse or banded (**sparse_precision** 0.63, **hierarchical** 0.68, **ar1** 0.70), ties across the dense elliptope middle, and loses under **residuals** and **factor** (1.03–1.09). Dense low-rank structure is the worst case for sparsity.

A smaller companion audit: with oracle shrinkage intensities, the constant-correlation target of Ledoit and Wolf [2003] beats their scaled-identity target [Ledoit and Wolf, 2004] on every draw under **sparse_precision** (median loss ratio 0.62) and loses on nineteen of twenty draws under **residuals**. Even the choice between two targets from the same authors is ensemble-relative.

A regime check guards against the $T = 2n$ coincidence: at $T = 120$ the intensity spread persists (0.04 to 0.84) and clipping still helps only **sparse_precision** and **hierarchical** (script beside the paper).

The remaining four audits concern inference: tests, certification rules, and reported dimensions. Their currency is rates and counts rather than loss ratios (Table 2).

Evolutionary biology. Cheverud [1996] declares two genetic covariance matrices similar when their mean response-vector correlation under common random selection gradients exceeds the null for independent random vectors. We apply the test to independent pairs drawn from the same ensemble. This is a population property with no sampling noise, so a sound test should declare similarity about 5% of the time. It does so under **residuals**, **factor**, and **archakov_hansen** (0%), and declares independent matrices similar 100% of the time under eleven of the fifteen ensembles. Under **sparse_precision** the mean skewers correlation between independent matrices is 0.97, because near-diagonal matrices respond to any skewer with the skewer itself. Rohlf [2017] raised this objection in general terms; the ensembles make it quantitative.

ensemble	HRP/MV (fin.)	clip/raw (econo.)	near/full (geost.)	PA count (psych.)	taper/raw (assim.)	load/raw (sig. proc.)	gl/LW (graph.)
animals	8.4	1.43	5.5	3.3	1.66	32.8	1.00
arl	0.98	1.20	1.00	4.5	0.53	0.48	0.70
archakov_hansen	71.8	1.47	131	4.7	1.41	20.5	1.00
block_equicorr	1.09	1.00	1.04	3.0	1.00	0.55	0.78
factor	12.3	1.38	1.06	2.2	1.69	0.94	1.09
hierarchical	1.00	0.76	1.02	2.3	0.82	0.45	0.68
kernel	1.51	1.46	5.7	4.1	1.46	136	1.00
lkj	2.27	1.22	3.1	6.5	0.74	0.87	0.97
onion	2.83	1.29	9.0	7.3	0.78	3.3	1.00
residuals	5.36	1.44	4.8	4.7	1.71	0.62	1.03
sparse_precision	0.76	0.30	1.00	1.8	0.50	0.48	0.63
spectrum	1.11	0.94	1.32	5.8	0.66	0.51	0.99
vine	2.68	1.29	11.9	7.0	0.81	4.1	1.00
walk	1.23	1.11	$\sim 10^{8.7}$	3.7	1.14	$\sim 10^{7.9}$	1.00
wishart	1.19	1.04	1.50	6.5	0.66	0.51	0.96

Table 1: Seven audited claims about estimators and constructions; all entries are medians over twenty seeded draws. Column 1: out-of-sample true-variance ratio, HRP over long-only minimum variance (below one: HRP wins). Column 2: Frobenius-loss ratio, Marchenko–Pastur-clipped over raw sample correlation (below one: cleaning helps). Column 3: population conditional-variance ratio, five best predictors over all (near one: screening holds). Column 4: mean components retained by parallel analysis. Column 5: Frobenius-loss ratio, Gaspari–Cohn-tapered over raw sample covariance at $N = 20$ members (below one: localization helps). Column 6: true beamformer output power, diagonally loaded over unloaded at $T = 40$ snapshots (below one: loading helps). Column 7: precision-matrix Frobenius-loss ratio, cross-validated graphical lasso over inverted Ledoit–Wolf (below one: the sparsity prior wins). Each column’s verdict flips across rows.

Climatology. North et al. [1982] declare a leading EOF trustworthy when its eigenvalue sampling error $\lambda\sqrt{2/T}$ is smaller than the gap to the next eigenvalue. We score the false-reassurance rate: among draws the rule declares separated, how often the leading sample eigenvector is mostly wrong, meaning squared alignment with the truth below one half. The rate is 0% under six ensembles, 57% under `wishart`, and 100% under `sparse_precision`, which declares separation on only 12% of draws but is always wrong when it does. The rule is a first-order perturbation statement: excellent for spiked spectra, unreliable exactly where spectra are quasi-degenerate, and silent about which case obtains.

Ecology. Guillot and Rousset [2013] report that the Mantel test [Mantel, 1967] rejects independence between distance matrices at 25–55% under spatially autocorrelated fields, at nominal 5%. Our audit puts the original claim and the critique on the same footing: two independent site variables drawn with covariance C , Euclidean distance matrices, 499 label permutations. The realized type-I error is the nominal 5% under `onion` and `vine`, and 37–40% under `residuals`, `factor`, `kernel`, and `animals`. This reproduces the published inflation under locality-and-factor structure and shows calibration under dense ellipsope measures. Neither “the Mantel test is broken” nor “the Mantel test is fine” is an ensemble-free sentence.

ensemble	skewers false sim.	North rule false reass.	Mantel type-I	n_D ratio est/true
animals	10%	0%	40%	0.89
ar1	100%	20%	13%	0.54
archakov_hansen	0%	9%	28%	0.78
block_equicorr	100%	0%	15%	0.77
factor	0%	0%	38%	0.90
hierarchical	100%	0%	6%	0.78
kernel	100%	3%	39%	0.88
lkj	100%	22%	11%	0.58
onion	100%	33%	5%	0.57
residuals	0%	0%	37%	0.79
sparse_precision	100%	100%	7%	0.42
spectrum	100%	10%	9%	0.52
vine	100%	33%	5%	0.55
walk	100%	0%	8%	0.83
wishart	100%	57%	7%	0.47

Table 2: Four audited claims about inference. Column 1: share of independent matrix pairs that random skewers declares similar (sound: $\approx 5\%$). Column 2: among draws North’s rule declares separated, share whose leading EOF is mostly wrong (squared alignment below $\frac{1}{2}$). Column 3: realized type-I error of the Mantel test at nominal 5%. Column 4: estimated over true effective dimensions of a covariance matrix at $T = 15$ (honest: near one).

Quantitative genetics. Kirkpatrick [2009] reports effective dimension counts $n_D = \sum_i \lambda_i / \lambda_{\max}$ of roughly one to two for published genetic covariance matrices, read as evolutionary constraint. At the low precision typical of breeding designs, $T = 15$ here, the estimated n_D deflates the truth by an ensemble-dependent factor. A true n_D of 12.3 under **sparse_precision** is reported as 5.2 (ratio 0.42), while **factor** and **animals** are nearly honest (0.90, 0.89) because their truth is already low-dimensional. Across all fifteen ensembles the estimate never exceeds 5.2, whatever the truth. A low reported n_D cannot by itself distinguish low-dimensional biology from a noisy estimate of diffuse structure.

7 One leaderboard per measure

The audits test one claim at a time; the same harness ranks whole libraries. We fit the six covariance estimators of scikit-learn and the eleven batch-applicable estimators of **precise** to the same twenty seeded panels per ensemble and rank them by median relative Frobenius loss against the generating covariance (Figure 2). Five estimators take first place somewhere: Ledoit–Wolf on four ensembles, OAS and fixed shrinkage and the graphical lasso on three each, and the raw sample covariance on **animals** and **kernel**, the two ensembles the intensity table already marked as wanting almost no shrinkage. The graphical lasso is first on the three precision-structured rows and tenth or worse under **animals**, **residuals**, and **factor**. The **precise** estimators are built for time-varying covariance and pay for that design on static panels, filling the lower ranks. A library default is a bet on a measure.

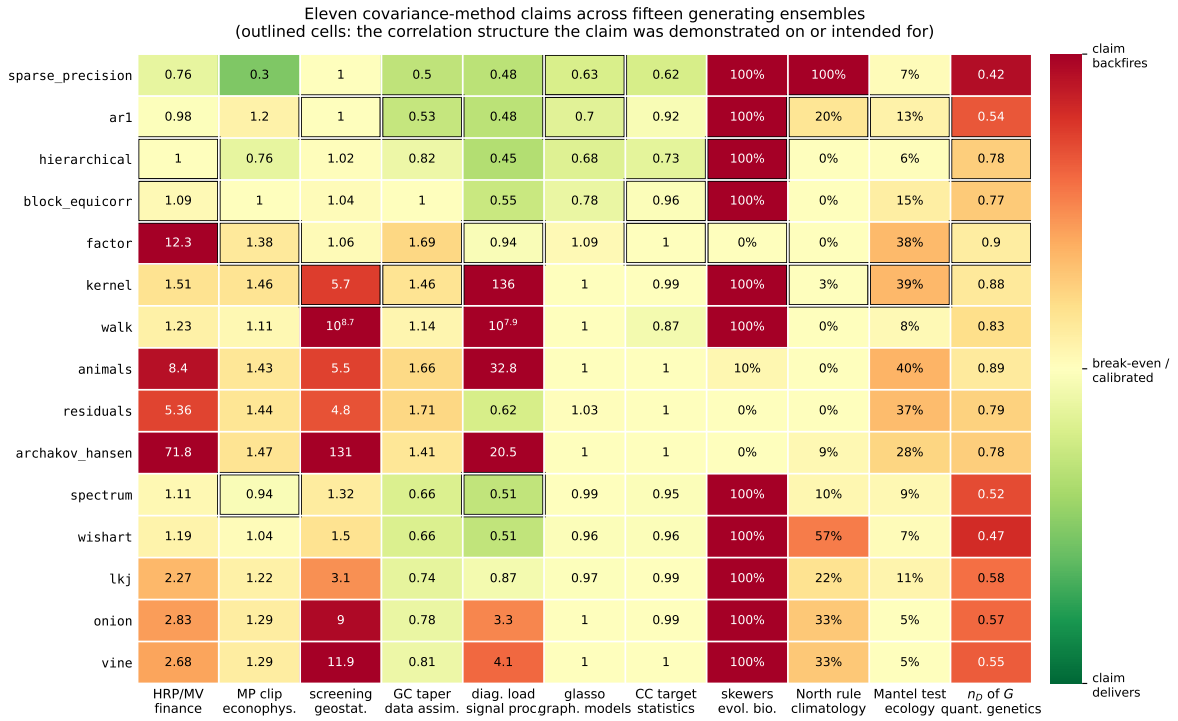


Figure 1: The audit matrix: eleven claims (columns, with their fields) against fifteen generating ensembles (rows, ordered roughly from structured to dense). Green: the claim delivers under that measure. Yellow: break-even or calibrated. Red: the claim backfires or the test is broken. Cell entries are the medians and rates of Tables 1 and 2. Outlined cells mark each claim’s home turf, the correlation structure it was demonstrated on or intended for.

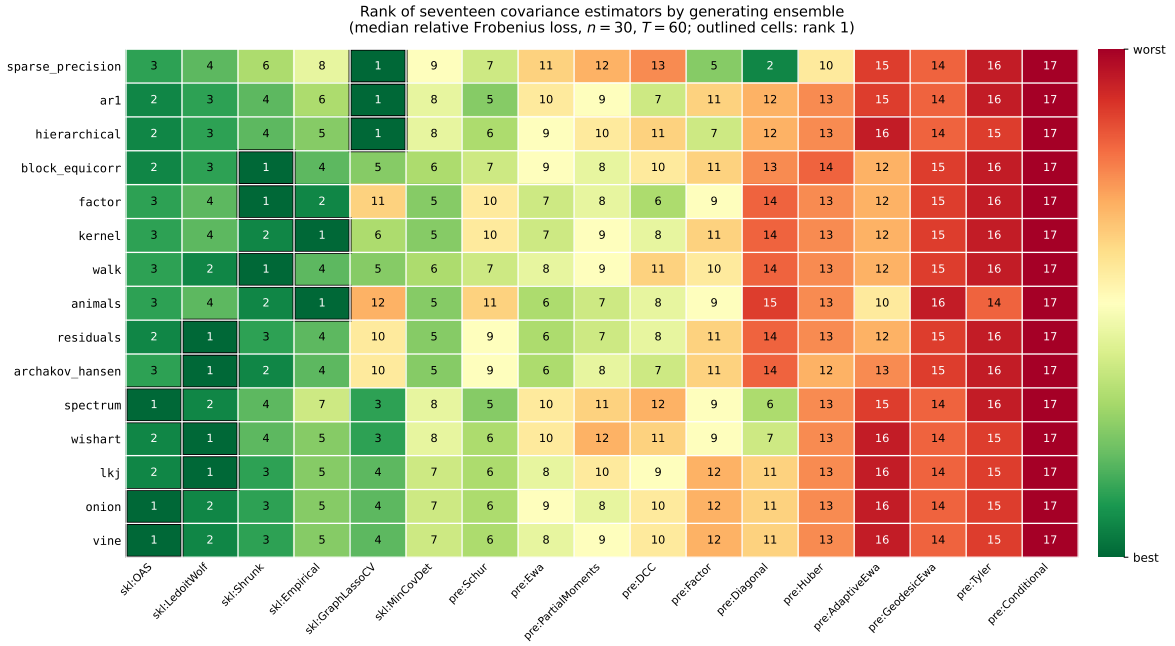


Figure 2: The leaderboard matrix: seventeen covariance estimators, six from scikit-learn and eleven from **precise**, ranked within each generating ensemble by median relative Frobenius loss at $n = 30$, $T = 60$, twenty seeded draws per ensemble. Rank-one cells are outlined; columns are ordered by mean rank.

8 Scope

The audits measure the ensemble-sensitivity of a claim, not the care of its authors. Stein proved that screening depends on smoothness; Bickel and Levina [2008] state the bandable class their banding rates require; López de Prado demonstrated HRP on equity covariances and said so; the data-assimilation literature has always tied localization to physical distance. Where a paper states its generating law, the audit maps the claim’s domain of validity onto named coordinates and adds nothing.

The exposure arises downstream. Verdicts certified under one measure are quoted, cited, and defaulted into software as if they were measure-free, and the second-hand version of a scoped claim is usually unscoped. Naming the ensemble gives the scope a form that survives citation.

9 The docket

The pattern across Figure 1 is not random: each field’s signature claim performs best on the ensembles that resemble the data the field grew up on, with instructive exceptions where a home-turf cell fails, as when screening fails inside the kernel ensemble. `sparse_precision` is kind to hierarchical bisection, eigenvalue cleaning, localization, and the graphical lasso at once, yet breaks random skewers and North’s rule. The dense ellipsope samplers (`lkj`, `onion`, `vine`) are kind to almost nothing except the Mantel test, and they are the most common default test bed. The same mechanism recurs under different names: the hedge that unconstrained minimum variance exploits is the null-steering that diagonal loading erases.

A survey of prominent covariance-method claims across further fields yields a docket for the next round, kept beside this paper. Sorting it teaches a taxonomy. Claims that impose structure on the truth are ensemble-relative; theorems about Wishart sampling noise hold for every true covariance and cannot flip.

The structural claims are the audit targets. Paz and Sánchez [2015] taper clustering-statistic covariances and presume banded truth; Padmanabhan et al. [2016] estimate sparse cosmological precisions and presume a sparse graph. Zhang and Horvath [2005] select gene-network powers by a scale-free-topology criterion that is itself a statement about the generating measure of gene correlation matrices. Smith et al. [2011] demonstrated the win of partial over full correlation for network detection on sparse simulated networks. Norberg et al. [2009] judged jackknife covariances for galaxy surveys, and the verdict hinges on cross-region correlation structure.

A second strand turns on spectral shape. Rule N [Preisendorfer and Mobley, 1988] truncates EOFs through the eigenvalue spacing law. Ledoit and Wolf [2012] derive the edge of nonlinear over linear shrinkage from spectral dispersion; Fan et al. [2013] build POET on pervasive factors plus sparse residuals; Frank and Friedman [1993] show the PLS-versus-PCR contest turning on how the coefficient vector aligns with the predictor eigenstructure. Each carries a predicted flip.

The invariant theorems serve as null controls. Hartlap et al. [2007] debias the inverse sample covariance exactly, for every true covariance; Dodelson and Schneider [2013] derive a parameter-variance inflation that depends only on counts; Sellentin and Heavens [2016] marginalize the estimated covariance exactly under Wishart sampling; Lowry et al. [1992] prove the MEWMA chart depends on a known covariance only through a noncentrality parameter. An audit

pipeline that finds ensemble-dependence in the Hartlap factor has a bug, so we ran that control: with one fixed covariance per ensemble and four hundred panels at $(n, T) = (30, 60)$, the trace of the raw average inverse sample covariance overshoots the true inverse trace by a factor of 2.07 to 2.13 under every ensemble, against the Wishart factor $(T - 1)/(T - n - 2) = 2.11$, and the corrected ratio is 0.98 to 1.01 everywhere. The pipeline reports no ensemble-dependence where none can exist. The most instructive failures are invariant theorems wrapped in ensemble-relative plug-in practice, as when those corrections are applied to shrunk or tapered estimates whose distribution is no longer Wishart.

No method in the tables dominates across the rows. Each owns the ensembles whose structure it assumes, and the matrix draws the borders. We leave to future work the remaining docket entries, beginning with the Hartlap factor as a pipeline control, the nonlinear-against-linear shrinkage comparison, and the WGCNA scale-free criterion.

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